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Multiple linear regression of metro ridership in the context of football events: a case study of Metro de Lisboa

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Abstract

Special events, such as sports games, concerts, and festivals, pose significant challenges for efficient operational management, leading to disruptions in public transport systems, particularly urban metros. While existing studies have focused on predicting metro passenger flows during sport events based on weather or opponent type variables, little is known about the impacts of competition type, team form, position in the competition, or game results on both game attendance and metro demand. This study examined inbound and outbound passenger flows at the station-level in the Lisbon metro during football games. The database comprised 186 events spanning from 2017 to 2022, including 15-minute mobility data from three metro stations serving two stadiums in the city. Multiple linear regression (MLR) was applied to estimate attendance games. Results revealed statistical differences in game attendance based on competition and opponent types, as well as team situation in the competition. Also, there was an increase in metro demand up to 3,600 passengers on event days compared to non-event days. The MLR models incorporating all the variables retained significantly more data information than other model variants for both teams. Some significant attendance model predictors were related to opponent and competition types, and team's situation before the games.

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1. Introduction and Research Objectives

Urban metro systems are low-carbon transport modes that are characterized by their punctuality, safety, and higher speeds (Lv et al., 2024), and are utilized as efficient measures to mitigate harmful effects of traffic congestion and fast

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urbanisation, as well as to benefit both urban development and social welfare (Ma et al., 2015).

Precise passenger flow prediction is essential in the planning, management, and operation of these systems (Hasan et al., 2013). Past research has mostly lie in predicting day-to-day normal passenger flows at different estimation levels: i) whole transit lines (Leng et al., 2013); ii) one station with passenger transfer flow (Zhou et al., 2020); and iii) one station with entrance and transfer flow (Sun et al., 2015). However, when it comes special events (e.g., sports, concerts, running races, etc.), the prediction of metro passenger flow burst turns into a challenging task because these events often lack substantial historical data, making it difficult for pattern recognition and machine learning models to accurately forecast passenger flows (Li et al., 2017).

Passenger flow prediction can be categorised into long-term and short-term prediction. The first category does not use daily traffic operations, which are particularly sensitive to external factors as special events. This makes long-term prediction less robust in capturing the deterministic and stochastic characteristics of subway ridership during special events (Zhang et al., 2013). Regarding the short-term, it shows as more reliable approach for passenger flow prediction (Li et al., 2017). The state-of-art studies in short-term prediction methods generally applied computational intelligence-based models and statistical approaches, including genetic algorithms (Abdulhai et al., 2002), support vector machine (Sun et al., 2015), neural networks (Fu et al., 2022), empirical mode decomposition (Wei and Chen, 2012) and time series (Li et al., 2023). However, most of existing studies only focus on analysing regular fluctuation of metro passenger flow to some extent (Pereira et al., 2015).

Special events, in particular football games, are relevant trip generators and have an aggregation effect on the passenger flow around metro stations before and after each event, which in turn is associated with both team historical data, game evolution and weather conditions. Understanding the spatial and temporal distributions of passenger flows in advance together to football event specificities requires attention to benefit route scheduling and station crowd regulation operations (Zhong et al., 2015).

A good body of literature has addressed the main variables affecting passenger flows during atypical conditions. Some of these studies made use of smartcard data, which are available in most of existing metro systems. Li et al. (2017) developed a multiscale radial basis function to estimate the irregular fluctuation of metro passenger flows prior a concert. Rodrigues et al. (2017) applied a Bayesian additive model with Gaussian process components to decompose the number of trips by routine commuting and by various sports and leisure events at two metro stations. They considered event specific variables, such as, day, duration, and specified as single or multi-day event. Chen et al. (2020) compared different generalised autoregressive conditional heteroscedasticity models to improve short-term predictions at two metro stations near a major sports centre, but they did not examine attendance. Qian et al. (2020) explored how weather and opponent type affect inbound and outbound passenger flows at three metro stations during sports events. They applied a gradient boosting decision tree prediction model based on 3 years of event data.

Other authors have used web data into the metro prediction passenger flow during special events. Pereira et al. (2015) used online information on sports, cultural and races events in bus and metro arrival prediction in Singapore network while Ni et al. (2017) found a moderate positive correlation between passenger flow and number of the hashtags from the Twitter users at one station near two major sport venues. Xue et al. (2022) adopted a multivariate disturbance-based hybrid deep neural network to predict outbound flow using data from 3 metro stations near stadiums.

While the literature review is rich concerning the passenger flow prediction at metro stations fusing historical data and social media, few studies have explored event specific features, such as competition type, team form, position in the competition or results during the distinct phases of the game on the attendance and number of event passengers. This paper focuses on analysing passenger flows at metro stations during sport events, specifically football games involving Sport Lisboa e Benfica (Benfica) and Sporting Clube de Portugal (Sporting), using daily mobility data from the Metro of Lisbon (Portugal). A multiple linear regression (MLR) approach was used to estimate the attendance related to these events.

The specific objective of the study is threefold: 1) to examine the impacts of opponent type, competition type, team form, position in the competition, day of the week, hour, 90-min and final result, temperature, presence of rain before event, and wind speed on the game attendance; 2) to evaluate origin-destination (O-D) passenger flow at three metro stations at the metro lines during the football events; and 3) to develop models able of predicting the attendance by team and test the most relevant game features that explained the variability in the data.

2. Methodology

This study uses football events variables to predict attendance values and mobility data to characterise passenger volumes at metro station level (Fig. 1). In the Section 2.1, we introduce the case study followed by the explanation of the dataset of football and metro data. The formulation of the MLR for the proposed problem is done in Section 2.2.

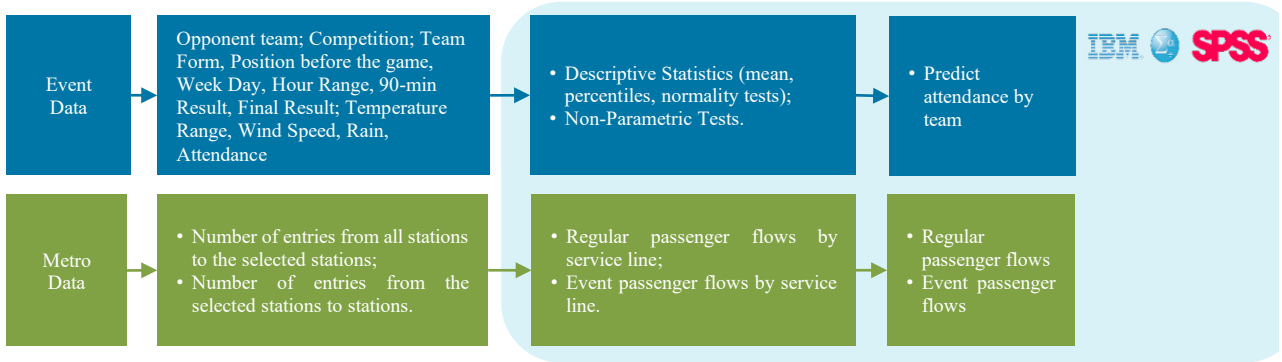


Fig. 1. Flowchart of the methodology.

2.1. Study Area and Dataset

The context of this study is the metropolitan area of Lisbon, Portugal. The city is covered by several public transport options, including buses, urban metro, trams, commuter trains and ferries, and funiculars. The Metro of Lisbon (ML) is a rapid transit system with 56 stations for 4 lines with a total extension of 44.5 km. In 2022, the annual ridership on the ML totalled 136.7 million passengers, with an available supply of almost 3,580 million seat-km (AMT, 2024).

The study focuses on two stadiums, namely Estádio da Luz (maximum attendance of 65,000) and Estádio de Alvalade (maximum attendance of 50,095), hosting the games of Benfica and Sporting, respectively (Fig. 2). Our research is centered on two metro stations (“Alto-dos Moinhos – AL” and “Colégio Militar” – CM) close to Estádio da Luz from blue line while the third station (“Campo Grande” – CG) serves the Estádio de Alvalade from yellow and green lines.

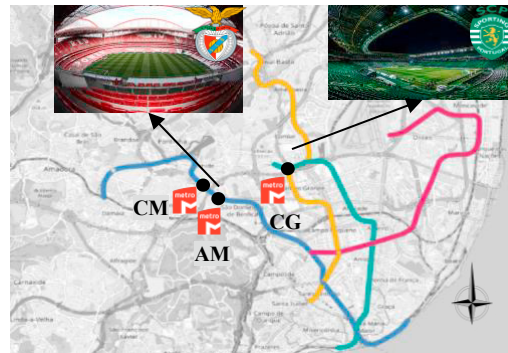


Fig. 2. Geographic distribution of the studied stadiums and metro stations. Image Source: (AMT, 2024)

From January 2017 to December 2022, a raw data set of 197 events was retrieved from different web sources. The periods between 2020 and 2021 were excluded due to COVID-19 restrictions. Lisbon area meteorological data were obtained from the Web on an hourly basis, either through screen scrapping or, when available, through the direct use of APIs. The classification of wind speed was based on Beaufort Wind Scale (Barua, 2019). Information regarding games and weather were collected from sources, namely: [zerozero.pt](https://www.zerozero.pt), [transfermarkt.co.uk](https://www.transfermarkt.co.uk), and [visualcrossing.com](https://www.visualcrossing.com). The first three season games were excluded from the analysis.

The mobility data used in this research correspond to the number of entries (ridership) from all stations to the selected stations and vice versa. Daily data during metro operating hours (6:30 a.m. to 01:00 a.m.) were aggregated into 15-minute intervals for each event day, along with the date seven days prior to the game, enabling a comparison of ridership between game event and non-event days. These mobility data were also aggregated by operating line of the ML system.

2.2. Model Formulation

This study seeks to develop an MLR (Multiple Linear Regression) model that allows to estimate the attendance number for the game where Benfica or Sporting teams are playing. For both cases match-related data, we developed an MLR model potentially involving the independent variables (regressors) listed in Tab. 1, where the variable Attendance will be considered dependent. Eleven variables were classified as nominal except for the game attendance (numerical).

Despite AFR and FR variables are not directly applicable with attendance values, they can be impact on mobility patterns after games, so that the research team keep them in the attendance predicting models.

The first step includes checking the correlation between variables. This is done based on the Spearman's correlation coefficient, which allows for examining the strength and direction of monotonic relationships, since most variables lie into ordinal/nominal data (Chatterjee and Simonoff, 2012). On a multivariate least squares fitting, considering N observations (which in our particular case, will be 97 observations for the Benfica dataset, and 89 for the Sporting) on a response Y (which in our case, will be the attendance number), with p independent variables (regressors) X_j , with $j = 1, \dots, p$, the MLR model can be given by (Chatterjee and Simonoff, 2012):

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} + \varepsilon_i \quad (1)$$

where $\beta_1, \dots, \beta_p$ are the estimates of the regression coefficients, β_0 is the model constant, and the ε_i is the error term.

Tab. 1 Description of the nominal variables

Feature	Description	Categories	Feature information
Opponent team (OT)	Type of Away Team	4	Other National; Top National; Other European; Top European
Competition (C)	Type of Competition	4	National League; National Cup; Champions; Europa League
Team Form (TF)	Number of team wins in the 3 prior games	4	0; 1; 2; 3
Position before Game (PBG)	Situation in the current competition	6	First Place; Second Place; Below Second Place; Group Stage; Knockout Phase; Qualifying
Week Day (WD)	Day of the game	7	Monday; Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday
Hour range (HR)	Range of hours of the game	3	11-18; 18-20; 20-22
90-min Result (AFR)	Result at the regular period	3	Win; Tie; Loose
Final result (FR)	Result in the end of the game	3	Win; Tie; Loose
Temperature range (TR)	Range of environmental temperatures (Celsius degrees) before the game (Boynton et al., 2019)	3	5-13; 13- 22; 22-35
Wind Speed (WS)	Range of wind speeds before the game (Barua, 2019)	4	Light Winds; Light Breeze; Gentle Breeze; Fresh Breeze
Rain (R)	Presence of rain before the game	2	Yes; No

To develop an MLR model and apply it correctly, certain assumptions underlying the statistical inference of the MLR model must be ensured, namely: i) there is no perfect multicollinearity between the independent variables; ii) homoscedasticity (variance of the residuals is constant); iii) multivariate normality (residuals are normally distributed); and iv) independence of observation (residuals are independent). Evaluating these assumptions is important to ensure that the MLR model is valid (Chatterjee and Simonoff, 2012). The model will be estimated through multivariate least squares and the performance of the estimated model is assessed based on the calculation of the R^2 .

3. Results

3.1. Football and Mobility Characteristics

The descriptive statistics of the attendance based on 186 events were as follows: i) Benfica (min = 16,454; P_{25} = 41,926, P_{50} = 52,224, P_{75} = 56,463 and max = 64,591); and ii) Sporting (min = 8,713; P_{25} = 27,671; P_{50} = 34,221, P_{75} = 42,706 and max = 49,339). Fig. 3 confirmed notable variations in attendance values based on the opponent and competition types, as confirmed by non-parametric Kruskal-Wallis tests that revealed significant differences (p-value < 0.001) in the median attendance for both teams. Also, the distributions of the Benfica attendance were not same across position (p-value < 0.001) and wind speed (p-value = 0.022) categories. For Sporting, the median attendance was statistically difference with respect to the presence of rain before the game (based on Wilcoxon rank-sum test and with a p-value = 0.044).

The analysis of the mobility data by line confirmed differences in passenger flows between event and non-event days considering a 3-h window of the game schedule (Tab. 2). These variations were clearer after games than before games as confirmed by the ratios between non-event and event values.

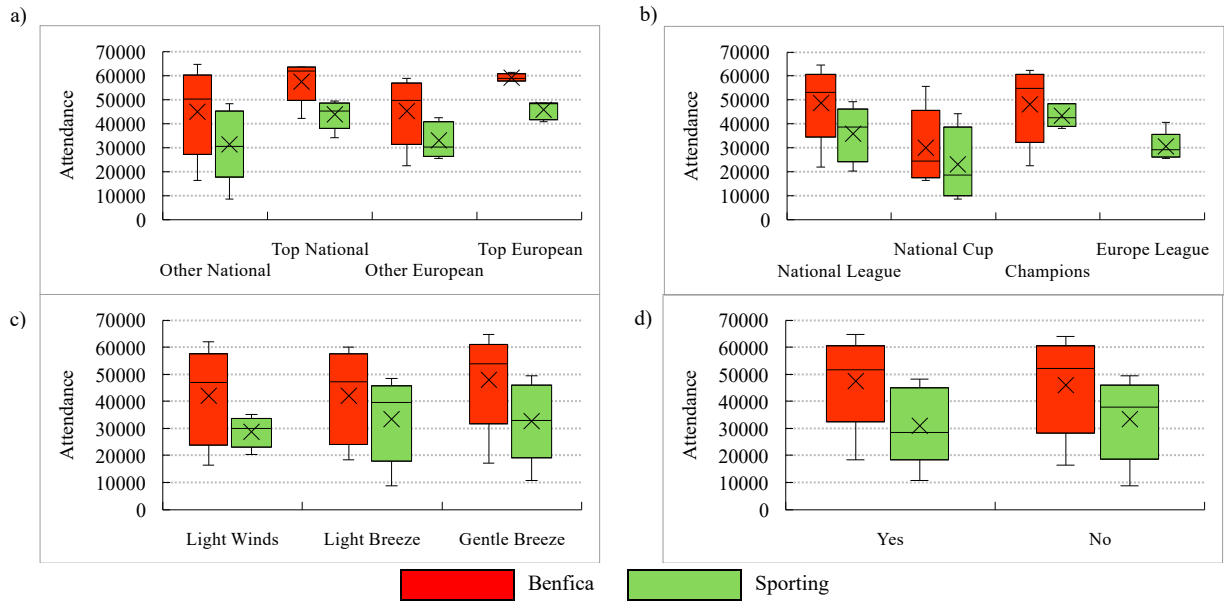


Fig. 3. Whisker plots of attendance values: a) Opponent; b) Competition; c) Wind Speed; and d) Rain. Notes: Categories with less than 6 observations were not plotted

Tab. 2 Game and mobility analysis at the line level for the two studied teams

Team	# Events	Average Attendance	Station	Period	Δ passenger flows between event and non-events by line				Ration between non-event and event flows (%)			
					Blue	Yellow	Green	Red	Blue	Yellow	Green	Red
Benfica	97	47,813 ($\pm 12,390$)	AM	3h Before Game	2,150 ($\pm 1,307$)	86 (± 37)	127 (± 46)	328 (± 363)	1.5	2.6	1.5	1.5
				3h After Game	1,380 (± 614)	200 (± 256)	243 (± 147)	539 (± 146)	1.9	1.5	1.6	1.9
			CM	3h Before Game	1,610 (± 785)	522 (± 603)	217 (± 96)	335 (± 142)	6.6	7.5	6.0	10.1
				3h After Game	1,443 ($\pm 1,125$)	96 (± 157)	139 (± 130)	262 (± 211)	9.2	6.8	8.0	16.6
Sporting	89	34,170 ($\pm 9,909$)	CG	3h Before Game	439 (± 172)	1,522 (± 473)	805 (± 330)	324 (± 123)	1.8	1.7	1.6	1.9
				3h After Game	261 (± 122)	1,648 (± 722)	1,370 (± 523)	370 (± 128)	4.2	3.6	4.0	4.1

Fig. 5 and Fig. 5 exemplify the fluctuation in station passenger flow that were aggregated into 15-min intervals in relation to the game schedule (the x-axis 0 is the beginning or end of the game). The research team chose 3-4 games of each team based on attendance percentiles. The pattern in the passenger flow between non-event and event was different before and after the games. The increase in the number of passengers occurred 1-2 hours prior the start of the game in both teams, with this increase being more spread out over time. Regarding the end of the game, the curve exhibited a flatter shape, with the peak difference in the number of passengers happening between 15-30 minutes after the game. The second peak in station passenger flow after Benfica games was due to a shopping mall near the stadium.

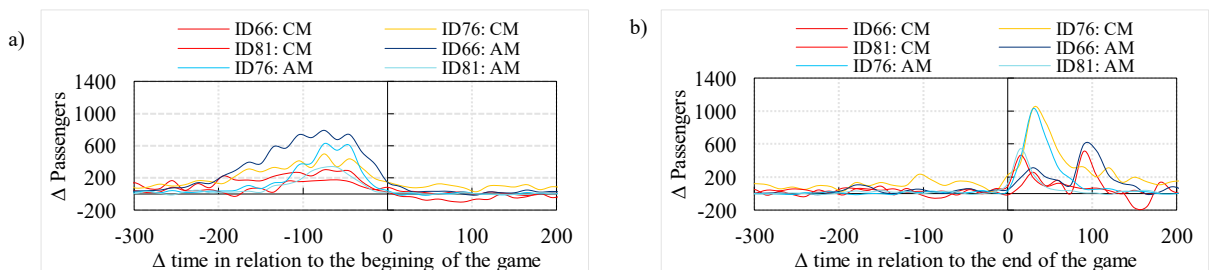


Fig. 4. Difference in passenger flows at the studied stations: a) Benfica – Before the Game; and b) Benfica – After the Game.

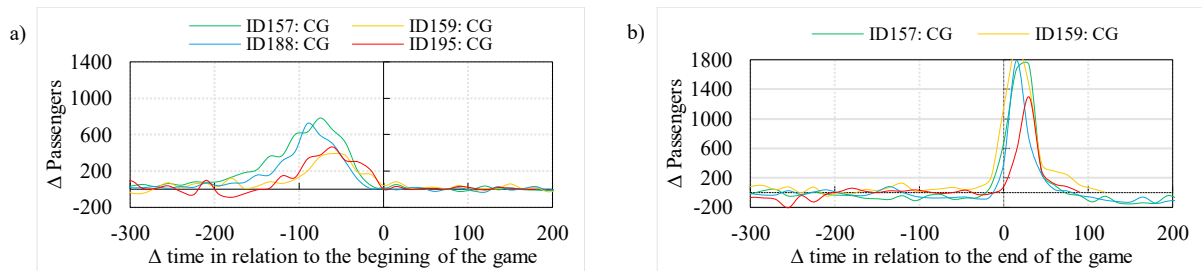


Fig. 5. Difference in passenger flows at the studied stations: a) Sporting – Before the Game; and b) Sporting – After the Game.

3.2. Prediction Modelling

This section presents the MLR model for estimating attendance by team. Given that the data includes qualitative or ordinal variables, dummy variables were generated as detailed in Tab. 3. A Spearman's correlation was run, and results suggest that there is a very strong, positive monotonic correlation between AFR and FR (Benfica: $r_s = 0.964$, p -value = 0, Sporting: 0.930), thus, only FR was retained and included in the variable set. The developed model excluded the variables C1 and C3 for the data related to Benfica, and OT3, C3, and PBG3 for Sporting case. Regarding the model fit statistics, the resulted R^2 is 68.6 % and 57.8 % for Benfica and Sporting, respectively. For a confidence level of 95%, given that F-test is 10.903 (6.658) (through ANOVA), with p -value 0 for both cases, we can ensure that at least one of the selected independent variables has a contribution to the developed models.

Tab. 3 Characteristics of MLR models

Feature	Classes	Dummy variables
Opponent	4	OT1 = 1 - Top European; OT2 = 1 - Top National; OT3 = 1 - Other European; 0 otherwise - Other National
Competition	4	C1 = 1 - Champions League; C2 = 1 - Europe League; C3 = 1 - National League; 0 otherwise - National Cup
Position	6	PBG1 = 1 - First Place; PBG2 = 1 - Second Place; PBG3 = 1 - Below Second Place; PBG4 = 1 - Knockout Phase; PBG5 = 1 - Group Stage; 0 otherwise - Qualifying

The estimates of the model parameters, as well as the results obtained for the t-test (which makes it possible to check which coefficients are statistically different from zero) are shown Tab. 4 for both Benfica and Sporting cases. Looking at the significance of the t-test, we can conclude that, among the independent variables chosen for inclusion in the MLR model for Benfica, seven variables (including the constant parameter) were statistically significant at the 5% significance level, as indicated by the shaded cells. Significant predictors for Benfica include OT1, OT2 and OT3 (opponent type), and PBG1, PBG2 and PBG3 (position). Specifically, these variables show that playing against stronger opponents and being in higher positions (e.g., first, second) have a strong positive impact on attendance. Non-significant predictors include competition types like C2 (Europe League), and lower positions and meteorological related variables.

Regarding Sporting, significant predictors include OT1, OT2, C1 and C2 (opponent type and competition), and PBG4 and PBG5 (knockout and group stage positions). This indicates that attendance is positively influenced by stronger competitions and opponents, but negatively affected by being in the knockout or group stages. Interestingly, higher positions did not significantly affect attendance for Sporting at the significant level of the test.

It should be noted that the inclusion of all the above-mentioned variables were relevant and have a contribution to the variability explained by each model, thus, even though some showed to have no statistical significance, in fact, the resulting MLR models showed to retain much more data information than other model variants (tested, but not included).

Multicollinearity between independent variables can be checked through the correlation matrix between all independent variables; and as previously shown, the developed model already did not consider AFR, that showed to be correlated with FR. It can be concluded that there are no collinearity problems, which validates the assumption.

To confirm the normality of the residuals, we can use the results of the Kolmogorov-Smirnov test, since the sample has more than 30 observations. The E_{obs} is 0.053 (0.096) and the p -value is 0.200 (0.400), so we can conclude that the null hypothesis H_0 (i.e., the errors are normal) is not rejected. The assumption can be considered validated.

With respect to the homoscedasticity of residuals, we consider H_0 as the errors have constant variance and this can be checked by analysing the graph of residuals versus adjusted predicted values. It was also confirmed that the residuals are randomly distributed around 0, meaning that the assumption is valid, i.e., hypothesis H_0 is not rejected.

Finally, to check whether the residuals are independent, the Durbin-Watson (D-W) test is considered. Since the result of the d test statistic is 1.681 (1.545), this value is between 1.5 and 2.5, and it can be concluded that the residuals are independent, and therefore we should not reject H_0 , meaning that the assumption is also valid. Based on the above analysis it can be stated that both models are suitable and reliable for estimating the attendance for each case.

Tab. 4 Model parameters estimates

Team	MLR	Unstandardized Coefficients		Standardized Coefficients		t	Sig.
		β	Std. Error	β			
Benfica	(Constant)	17403.661	9354.615			1.860	0.066
	OT1	37973.836	4560.594	0.681		8.327	0.000
	OT2	14092.916	3164.697	0.315		4.453	0.000
	OT3	21591.817	3509.118	0.633		6.153	0.000
	C2	-228.075	5875.373	-0.003		-0.039	0.969
	TF	1696.494	1203.554	0.105		1.410	0.163
	PBG1	25213.962	7015.700	0.906		3.594	0.001
	PBG2	20268.414	7213.810	0.541		2.810	0.006
	PBG3	15774.809	7056.324	0.574		2.236	0.028
	PBG4	2070.358	6878.206	0.053		0.301	0.764
	PBG5	-1797.372	6208.891	-0.058		-0.289	0.773
	WD	375.372	616.703	0.058		0.609	0.544
	HR	-2148.356	1477.565	-0.107		-1.454	0.150
	FR	3266.504	1379.796	0.182		2.367	0.020
	R	-969.806	1848.430	-0.036		-0.525	0.601
	TR	2391.122	1370.514	0.122		1.745	0.085
	WS	373.405	1302.977	0.021		0.287	0.775
Sporting	(Constant)	26390.674	6861.940			3.846	0.000
	OT1	10298.106	4720.238	0.217		2.182	0.032
	OT2	14228.872	2867.342	0.435		4.962	0.000
	C1	21697.652	4236.382	0.593		5.122	0.000
	C2	10414.682	3231.383	0.334		3.223	0.002
	TF	922.452	849.817	0.086		1.085	0.281
	PBG1	2248.499	7403.295	0.024		0.304	0.762
	PBG2	621.101	2074.493	0.026		0.299	0.765
	PBG4	-15224.092	3571.420	-0.509		-4.263	0.000
	PBG5	-16750.186	3288.452	-0.683		-5.094	0.000
	WD	141.184	663.334	0.026		0.213	0.832
	HR	975.360	1340.416	0.059		0.728	0.469
	FR	-54.292	1192.107	-0.004		-0.046	0.964
	R	-1485.936	1802.672	-0.070		-0.824	0.412
	TR	380.028	1361.527	0.025		0.279	0.781
	WS	1191.436	1262.176	0.077		0.944	0.348

4. Conclusions

This study analysed how football game factors influence attendance and Lisbon Metro ridership. A multi-linear regression was applied to estimate attendance based on the characteristics of football games for each team. Results show opponent and competition types significantly impact attendance, while day, hour, and temperature do not.

The mobility data analysis revealed a higher variation in passenger volumes after games compared to before games, with ratios between non-event and event values reaching up to 10 before games and 17 after games. The difference curve between event and non-event passengers exhibited a flatter shape after games, with peak values occurring 15-30 minutes after their conclusion. All multi-linear regression models were deemed suitable for attendance prediction and effectively retained football game information using all variables compared to models with fewer predictors.

The main contribution of the study is the analysis and inclusion of multiple soccer variables (some of them not explored before) and their significance on team attendance, particularly opponent and competition types, and team situation prior games. The proposed models at team level can have practical implications for metro operators to manage subway system schedules and resources and implement appropriate countermeasures in advance to deal with unexpected volatility associated with host large-scale events and concurrently enhance service efficiency and passenger comfort.

Despite the promising findings, this paper has some limitations, particularly regarding the size of the dataset, which led to a low frequency of certain group categories (e.g., competition types, positions). Additionally, the study lacked an analysis of external factors, such as the presence of nearby bus stops or commercial and service developments around metro stations, which could influence the estimation of event passengers.

Future work will focus on predicting passenger flows before and after soccer games using the data collected here. The comparison of the performance of various modelling approaches for passenger flow prediction, including time-series analysis, support vector machines, genetic algorithms, and neural networks will be also examined. The next research steps will also involve incorporating additional variables, such as vehicle load, train frequency, presence of multiple events, effect of 90-min result and Final result, nearby bus trip data, and route preferences. This will enable a more in-depth analysis of sudden passenger flows and facilitate the development of a detailed system to distinguish between irregular passengers attending large-scale events and regular passengers.

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